

Model Reference Adaptive Control of an Electrohydraulic Actuator: Input-Matrix Structure Analysis, Output-Feedback Adaptation, and Practical Stability

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Abstract

This paper presents a model reference adaptive control (MRAC) framework for a nonlinear electrohydraulic actuator (EHA) comprising a Moog G761 servo valve, a double-rod hydraulic cylinder, and a LuGre friction model. A scaled three-state representation in [mm, mm/s, bar] is adopted to improve numerical conditioning. The reference model is derived by pole-placement on the nominal closed-loop linearisation, so the model poles are physically consistent with the actuator dynamics.

A central structural observation is established analytically: because the scaled input matrix takes the form $\bar{B}_{mA} = [0 \ 0 \ b_3]^\top$, the full Lyapunov adaptation term $\mathbf{e}^\top P \bar{B}_{mA}$ is dominated by the pressure error state e_3 through the coefficient $P_{33}/P_{13} \approx 771$. During stiction phases e_3 diverges from the reference model while e_1 (position error) remains physically meaningful, causing the standard adaptation law to fail. We show that the position-channel projection $\varphi = e_1 (C P \bar{B}_{mA})$, where $C = [1, 0, 0]$ is the output matrix, is mathematically equivalent to *output-feedback MRAC* applied to the measurable position error: $\varphi = e_y (C P \bar{B})$.

A rigorous analysis then demonstrates that the condition required for a classical Uniformly Ultimately Bounded (UUB) proof via Young's inequality, $b_3(P_{23} + P_{33}) < \lambda_{\min}(Q)/(2 \|\tilde{L}\|_{\max})$, is *structurally infeasible* for this system class ($c_\delta = 100,337$ vs. a threshold of 5×10^{-6}). This represents a known theoretical limitation of output-channel MRAC in hydraulic systems with relative degree three. A norm-projection operator is incorporated to guarantee parameter boundedness by design, and σ -modification provides practical stability.

Simulation on the nonlinear Simulink model shows: steady-state position RMS error 0.45 mm (3 % of amplitude), convergence in ≈ 5 s, and a $6\times$ improvement over the unmodified adaptive law. Experimental validation on the physical test bench is planned as future work.

Keywords: electrohydraulic actuator; model reference adaptive control; output-feedback MRAC; σ -modification; LuGre friction; practical stability.

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1 Introduction

Electrohydraulic actuators (EHAs) are widely deployed in aerospace actuation, industrial robotics, and heavy machinery because of their high power-to-weight ratio, rapid dynamic response, and capacity for large force generation [1, 2]. Precise position tracking is challenged by nonlinear valve flow dynamics, hydraulic fluid compressibility, and friction at the piston–cylinder interface [3].

Model Reference Adaptive Control (MRAC) has attracted sustained research interest for mechanical systems because it does not require exact plant parameters and can compensate for slow parameter variations in real time [4, 5]. Classical MRAC theory is well established for relative-degree-one, Strictly Positive Real (SPR) systems; its extension to hydraulic systems with relative degree three and non-minimum-phase zeros presents both theoretical and practical challenges [6, 7].

A specific structural difficulty has received limited explicit attention in the literature. In the standard three-state EHA representation the input matrix satisfies $B = [0 \ 0 \ b_3]^\top$, so the gearbox input (valve current) directly excites only the pressure dynamics. The Lyapunov adaptation term $\mathbf{e}^\top PB$ therefore reduces to $b_3(P_{13}e_1 + P_{23}e_2 + P_{33}e_3)$, with $P_{33}/P_{13} \approx 771$ for the standard $Q = I$ solution. During stiction phases the plant differential pressure z_3 accumulates far above the reference model value $z_{m3} \approx 0$, making e_3 large and negative and driving the adaptation in the wrong direction.

The main contributions of this paper are:

1. A formal analysis of the B-matrix dominance problem in EHA MRAC, including an LMI-based verification (YALMIP/MOSEK) that the dominance cannot be eliminated by Q -matrix tuning or P -matrix optimisation.
2. The identification that the position-channel adaptation projection $\varphi = e_y(CPB)$ is mathematically equivalent to *output-feedback MRAC* on the measurable position signal, providing a theoretically grounded reframing of what was previously described as a heuristic “trick”.
3. A complete analysis of the fundamental theoretical limitation: a necessary condition for classical UUB is shown to be structurally infeasible for this system class, and three Propositions (parameter boundedness, theoretical limitation, practical stability) are stated precisely.
4. A norm-projection operator that guarantees parameter boundedness by design, combined with σ -modification for practical stability.
5. Simulation results on a high-fidelity nonlinear model (LuGre friction, valve dead-zone, pressure dynamics) confirming sub-millimetre steady-state tracking.

2 System Modelling

2.1 Electrohydraulic Actuator

The test-bench actuator consists of a Moog G761 proportional servo valve connected to a double-rod hydraulic cylinder. The cylinder drives a load carriage with equivalent mass $m = 3.4 \text{ kg}$ and viscous damping $B_p = 38 \text{ N} \cdot \text{s/m}$. Physical parameters are listed in Table 1.

Table 1: Electrohydraulic actuator parameters.

Parameter	Symbol	Value	Unit
Piston diameter	d_p	34	mm
Rod diameter	d_r	30	mm
Effective area	A_c	2.011×10^{-4}	m^2
Equivalent mass	m	3.4	kg
Viscous damping	B_p	38	$\text{N} \cdot \text{s/m}$
Bulk modulus	β_e	1.6×10^9	Pa
Effective volume	V_e	6.886×10^{-5}	m^3
Internal leakage	C_{ip}	3.993×10^{-13}	$\text{m}^3/(\text{s} \cdot \text{Pa})$
Supply pressure	P_s	199.85	bar
Flow-gain coeff.	K_q	1.100×10^{-2}	$\text{m}^3/(\text{s} \cdot \text{A})$
Pressure-flow coeff.	K_c	3.963×10^{-13}	$\text{m}^3/(\text{s} \cdot \text{Pa})$

Valve coefficients K_q and K_c are obtained from static bench measurements at reference pressures $\hat{P}_{s,fg}$ and $\hat{P}_{s,pg}$, scaled to the operating supply pressure $P_{s,\text{net}} = 195.4 \text{ bar}$ via orifice-law corrections:

$$K_q = \hat{K}_q \sqrt{\frac{P_{s,\text{net}}}{\hat{P}_{s,fg}}}, \quad K_c = \hat{K}_c \sqrt{\frac{\hat{P}_{s,pg}}{P_{s,\text{net}}}}. \quad (1)$$

2.2 Nonlinear Plant Model

The nonlinear plant comprises three subsystems: (a) a Moog G761 servo valve modelled as a first-order transfer function with a dead-zone nonlinearity (± 0.02 normalised); (b) nonlinear pressure dynamics with compressibility and internal leakage; and (c) cylinder mechanics with a LuGre friction model capturing stick-slip behaviour.

2.3 Linearised Scaled State-Space Representation

Let the physical states be $x = [x_c, \dot{x}_c, P_L]^\top$ where x_c is piston position, $\dot{x}_c$ velocity, and $P_L = P_A - P_B$ differential load pressure. The linearised continuous-time system is

$\dot{x} = Ax + Bu$, $y = Cx$, with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -B_p/m & A_c/m \\ 0 & -4\beta_e A_c/V_e & -4\beta_e(K_c + C_{ip})/V_e \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ (4\beta_e/V_e)K_q \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \quad (2)$$

To improve numerical conditioning, the coordinate transformation $T = \text{diag}(10^{-3}, 10^{-3}, 10^5)$ maps the states to $\mathbf{z} = T^{-1}x \in [\text{mm}, \text{mm/s}, \text{bar}]$. Control input u (valve current, A) is further scaled to milliamperes by a factor of 10^{-3} . The scaled matrices are:

$$\bar{A} = T^{-1}AT, \quad \bar{B}_{mA} = 10^{-3}T^{-1}B = \begin{bmatrix} 0 \\ 0 \\ b_3 \end{bmatrix}, \quad b_3 = 10,225.6. \quad (3)$$

The critical structural property $\bar{B}_{mA} = [0 \ 0 \ b_3]^\top$ is a consequence of the valve–pressure–position signal chain and is independent of the coordinate scaling chosen. It holds for all double-rod hydraulic cylinder systems of this type.

3 MRAC Design

3.1 Reference Model

The reference model is selected as the nominal closed-loop system obtained by state-feedback gain K_m on the scaled linearisation:

$$A_m = \bar{A} - \bar{B}_{mA}K_m, \quad \dot{\mathbf{z}}_m = A_m\mathbf{z}_m + B_m r, \quad (4)$$

where K_m is computed by pole placement targeting $\omega_n = 40 \text{ rad/s}$, $\zeta = 0.707$ for the dominant complex pair and a fast real pole at -150 rad/s :

$$p_{1,2} = -28.28 \pm 28.29j, \quad p_3 = -150, \quad K_m = [0.0040, -0.0181, 0.0119]. \quad (5)$$

The input matrix $B_m = \bar{B}_{mA}/\text{DC_gain}$ is scaled so that the position DC gain from reference r to output z_{m1} equals unity. Numerically, $\text{DC_gain} = C(-A_m)^{-1}\bar{B}_{mA} = 251.96 \text{ mm/mA}$ and $|G_m(j \cdot 2\pi \cdot 0.2)| = 1.000$.

The Lyapunov equation $A_m^\top P + PA_m = -Q$ with $Q = I_3$ yields:

$$P = \begin{bmatrix} 13.48 & 0.066 & 0.012 \\ 0.066 & 0.013 & 0.314 \\ 0.012 & 0.314 & 9.499 \end{bmatrix}. \quad (6)$$

3.2 Control Law

The MRAC control law is:

$$u = -\hat{L}^\top \mathbf{z} + \hat{M} r, \quad (7)$$

where $\hat{L} \in \mathbb{R}^3$ and $\hat{M} \in \mathbb{R}$ are time-varying adaptive gains initialised at $\hat{L}(0) = -K_m$ and $\hat{M}(0) = 1/\text{DC_gain}$.

3.3 B-Matrix Dominance in the Lyapunov Adaptation Term

The standard MRAC Lyapunov adaptation term is:

$$\mathbf{e}^\top P \bar{B}_{mA} = b_3 (P_{13} e_1 + P_{23} e_2 + P_{33} e_3). \quad (8)$$

From (6): $P_{13} = 0.012$, $P_{23} = 0.314$, $P_{33} = 9.499$, so $P_{33}/P_{13} = 771$ and $P_{23}/P_{13} = 25$. The pressure-error channel is dominant by two orders of magnitude.

During stiction phases the plant differential pressure z_3 accumulates while the reference model predicts $z_{m3} \approx 0$, so $e_3 = z_{m3} - z_3 < 0$ and large. With $b_3 P_{33} |e_3| \gg b_3 P_{13} |e_1|$, the scalar term in (8) becomes negative, driving $\dot{\hat{M}} < 0$ and collapsing the feedforward gain regardless of the position tracking error.

An LMI search (YALMIP/MOSEK) minimising $b_3(P_{23} + P_{33})$ over symmetric positive-definite P satisfying $A_m^\top P + P A_m \preceq -\varepsilon I$ reduces the ratio P_{33}/P_{13} from 771 to 8.89, but $c_\delta = b_3(P_{23} + P_{33})$ only decreases from 100,337 to 92,025—a 8% reduction that is insufficient to resolve the dominance in practice.

3.4 Output-Feedback Adaptation Projection

Observe from (3) that the measurable output is $y = C\mathbf{z} = e_1$ (position), and $CPB = P_{13}b_3$. We define the projected scalar driving term as:

$$\boxed{\varphi = e_y \cdot (C P \bar{B}_{mA}) = e_1 \cdot P_{13}b_3 = 125.98 e_1.} \quad (9)$$

Key observation. The projection (9) is not an ad-hoc simplification but is the *exact output-feedback MRAC* adaptation term. In output-feedback MRAC the adaptation driving signal is $\varphi = e_y(CPB)$, where $e_y = y_m - y$ is the measurable output error. Since $C = [1, 0, 0]$ selects position, and $\bar{B}_{mA} = [0 \ 0 \ b_3]^\top$ makes only the third column of P relevant, $CPB = P_{13}b_3$. Hence (9) is both physically interpretable (adaptation is driven exclusively by the measurable position error) and theoretically grounded.

The adaptation laws are:

$$\dot{\hat{L}} = \text{Proj}_{\|\hat{L}\| \leq L_{\max}} \left(-\Gamma_L \varphi \mathbf{z} - \sigma_L \hat{L} \right), \quad (10)$$

$$\dot{\hat{M}} = \text{Proj}_{|\hat{M}| \leq M_{\max}} \left(+\Gamma_R \varphi r - \sigma_M \hat{M} \right). \quad (11)$$

Design parameters: $\Gamma_L = 10^{-6}$, $\Gamma_R = 10^{-5}$, $\sigma_L = \sigma_M = 0.001$, $L_{\max} = 0.123$, $M_{\max} = 0.120$.

4 Stability Analysis

4.1 Closed-Loop Error Dynamics

Define $\tilde{L} = \hat{L} + K_m$ and $\tilde{M} = \hat{M} - K_r$ ($K_r = 1/\text{DC_gain}$). Substituting the control law into the plant and using $A_m = \bar{A} - \bar{B}_{mA}K_m$:

$$\dot{\mathbf{e}} = A_m \mathbf{e} - \bar{B}_{mA}(\tilde{L}^\top \mathbf{z} - \tilde{M} r). \quad (12)$$

4.2 Lyapunov Decomposition

Select the composite Lyapunov candidate:

$$V = \mathbf{e}^\top P \mathbf{e} + \frac{1}{\Gamma_L} \|\tilde{L}\|^2 + \frac{1}{\Gamma_R} \tilde{M}^2. \quad (13)$$

Since $\bar{B}_{mA} = [0 \ 0 \ b_3]^\top$, write $\mathbf{e}^\top P \bar{B}_{mA} = \varphi + \delta$ where $\varphi = e_1 P_{13} b_3$ (projected term) and $\delta = b_3(P_{23}e_2 + P_{33}e_3)$ (residual). After substituting (10)–(11) and noting the projection property $\tilde{L}^\top \text{Proj}(\tau, \hat{L}) \leq \tilde{L}^\top \tau$, the φ -terms cancel exactly and:

$$\dot{V} \leq -\mathbf{e}^\top Q \mathbf{e} + 2\delta(\tilde{M} r - \tilde{L}^\top \mathbf{z}) - \frac{\sigma_L}{2\Gamma_L} \|\tilde{L}\|^2 - \frac{\sigma_M}{2\Gamma_R} \tilde{M}^2 + C_\sigma, \quad (14)$$

where $C_\sigma = (\sigma_L/2\Gamma_L) \|K_m\|^2 + (\sigma_M/2\Gamma_R) K_r^2 = 0.244$ is a finite positive constant.

4.3 Circularity in the Perturbation Bound and Structural Limitation

The residual $\delta = b_3(P_{23}e_2 + P_{33}e_3)$ depends on the system states e_2, e_3 ; the term $\kappa = \tilde{M}r - \tilde{L}^\top \mathbf{z}$ depends on $\tilde{L}$, $\tilde{M}$, and $\mathbf{z} = \mathbf{z}_m - \mathbf{e}$. A naive application of Young's inequality to $2|\delta||\kappa|$ yields:

$$2|\delta||\kappa| \leq 2c_\delta \|\mathbf{e}\| \left[|\tilde{M}| r_{\max} + \|\tilde{L}\| (Z_m + \|\mathbf{e}\|) \right] = 2c_\delta \kappa_0 \|\mathbf{e}\| + 2c_\delta \|\tilde{L}\|_{\max} \|\mathbf{e}\|^2, \quad (15)$$

where $c_\delta = b_3(P_{23} + P_{33}) = 100,337$, $Z_m = \|G_m\|_\infty r_{\max} = 579$ mm, and $\kappa_0 = \|\tilde{M}\|_{\max} r_{\max} + \|\tilde{L}\|_{\max} Z_m = 85.9$. Substituting (15) into (14), the coefficient of $\|\mathbf{e}\|^2$ becomes:

$$\lambda_{\min}(Q) - 2c_\delta \|\tilde{L}\|_{\max} = 1 - 2 \times 100,337 \times 0.145 = -29,101 \ll 0. \quad (16)$$

Proposition 4.1 (Structural Infeasibility). *For a hydraulic system with $B = [0 \ 0 \ b_3]^\top$ and $Q = I$ in the Lyapunov equation, the condition $c_\delta < \lambda_{\min}(Q)/(2\|\tilde{L}\|_{\max})$ required for a classical UUB proof via (14)–(15) is structurally infeasible. In the present system:*

$$c_\delta = 100,337, \quad \frac{\lambda_{\min}(Q)}{2\|\tilde{L}\|_{\max}} \approx 5 \times 10^{-6}, \quad (17)$$

so the condition requires $\|\hat{L}\| < 5 \times 10^{-6}$ mA/mm, while the nominal gain satisfies $\|K_m\| = 0.022$ mA/mm. This is four orders of magnitude too large and cannot be resolved by any physical control design.

Remark 4.2. LMI-based minimisation of c_δ reduces it from 100,337 to 92,025 (8 % reduction) while maintaining $A_m^\top P + PA_m \prec 0$. The feasibility threshold remains unreachable. This is therefore a fundamental limitation of the Young’s-inequality-based UUB approach for this system class, not a consequence of a particular P -matrix choice.

4.4 Parameter Boundedness and Practical Stability

Despite Proposition 4.1, the following results hold rigorously.

Proposition 4.3 (Parameter Boundedness). *Under the projection laws (10)–(11), $\|\hat{L}(t)\| \leq L_{\max}$ and $|\hat{M}(t)| \leq M_{\max}$ for all $t \geq 0$, by construction of the projection operator.*

Proposition 4.4 (Practical Stability). *Under Proposition 4.3 with $\sigma > 0$, and for bounded reference r and bounded initial conditions, the adaptive gains $\hat{L}, \hat{M}$ remain bounded, the reference state $\mathbf{z}_m$ is bounded (since A_m is Hurwitz and r is bounded), and the position error satisfies $|e_1(t)| \leq \varepsilon_e$ for all large t , where ε_e is determined by $\sigma, \Gamma_L, \Gamma_R, L_{\max}$, and $M_{\max}$. Numerical simulation (Section 5) gives $\|\mathbf{e}_1\|_{\infty,ss} \leq 1.72$ mm.*

Remark 4.5. A complete UUB proof would require either (a) an alternative Lyapunov structure that does not require δ to be treated as an external perturbation, or (b) a different adaptation architecture that eliminates δ by design, such as a pressure inner-loop that drives $e_3 \rightarrow 0$ independently. Both constitute open problems for this system class.

5 Simulation Results

5.1 Setup

The nonlinear plant is implemented in MATLAB/Simulink R2023b with a fixed-step ode4 solver at 1 ms. Initial conditions are taken from experimental test-bench data: $P_A(0) \approx 71.3$ bar, $P_B(0) \approx 72.0$ bar, $\dot{x}_c(0) \approx -0.05$ mm/s. Reference trajectory: $r(t) = 15 \sin(2\pi \cdot 0.2 t)$ mm (amplitude 15 mm, 0.2 Hz). Simulation duration: 30 s.

5.2 Tracking Performance

Table 2 summarises position error statistics in successive time windows. The system enters steady-state tracking (sliding-window RMS below 1 mm) at approximately $t = 5$ s. The adaptive feedforward gain $\hat{M}$ converges from 3.97×10^{-3} to 2.38×10^{-2} mA/mm, reflecting the additional gain required to overcome LuGre static friction.

Table 2: Position tracking performance by time window.

Time window (s)	RMS error (mm)	Max error (mm)	Error / Amplitude (%)
0–5	6.40	11.49	42.7
5–10	1.99	6.22	13.3
10–20	0.44	1.52	2.9
20–30	0.45	1.72	3.0

5.3 Comparison Study

Table 3 compares five design variants. The standard full $\mathbf{e}^\top P B$ driving term without σ -modification diverges ($\hat{M} \rightarrow -9,257$). LMI-optimised P matrices reduce P_{33}/P_{13} but fail to prevent e_3 dominance. The proposed output-feedback projection with σ -modification achieves the best steady-state performance.

Table 3: Comparison of MRAC adaptation variants (30 s, 0.2 Hz, 15 mm reference).

Variant	c_δ	RMS $t > 20$ s (mm)	Converges?	UUB pr
Full $\mathbf{e}^\top P B$, no σ	100,337	32.7 (diverge)	No	No
LMI P (ratio 8.89), no σ	92,025	9.09	Partial	No
LMI P (ratio 1.18), no σ	92,025	4.76	Partial	No
Output-FB projection, no σ	—	0.44	Yes	No
Output-FB projection + σ + Proj (proposed)	—	0.45	Yes (≈ 5 s)	Prop. 4.3

5.4 σ -Modification Parameter Study

Table 4 reports steady-state RMS error and final $\hat{M}$ for $\sigma \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$. The value $\sigma = 0.001$ provides the best balance: RMS 0.17 mm and $\hat{M} \rightarrow 0.028$ mA/mm.

Table 4: Effect of σ on steady-state performance ($\Gamma_L = 10^{-6}$, $\Gamma_R = 10^{-5}$).

σ	RMS $t > 20$ s (mm)	$\hat{M}(30 \text{ s})$	Note
0.001	0.17	0.028	Optimal (selected)
0.005	0.43	0.021	Slight $\hat{M}$ bias
0.010	0.40	0.018	—
0.050	0.30	0.006	$\hat{M}$ suppressed
0.100	0.91	0.002	Performance degraded

6 Discussion

The output-feedback reframing (Section 3.4) clarifies why the e_1 -projection is not an ad-hoc remedy but rather a principled design choice: it drives adaptation via the only state that is directly observable and physically meaningful during stiction. The pressure error e_3 is large whenever the plant is in stiction because the actuator builds pressure without moving, while the linear reference model has no mechanism to replicate this behaviour.

The structural infeasibility result (Proposition 4.1) is of independent interest. It shows that for any hydraulic actuator with $B = [0 \ 0 \ b_3]^\top$ and closed-loop poles in the range used here, the Lyapunov-based UUB proof via Young’s inequality requires the adaptive gain to be smaller than the nominal pole-placement gain by four orders of magnitude—an impossibility. This is not a limitation of the particular plant or design parameters but follows from the algebraic relationship between b_3 , P_{33} , and $\lambda_{\min}(Q)$.

The projection operator resolves parameter drift and enables Proposition 4.3, while σ -modification drives adaptation toward the ideal gains, yielding Proposition 4.4. The combination achieves convergence six times faster than the unmodified law and maintains sub-millimetre accuracy after the adaptation transient.

Early-phase tracking error ($t < 5$ s, RMS ≈ 6.4 mm) reflects the stiction regime: $\hat{M}$ must grow from 3.97×10^{-3} to ≈ 0.025 mA/mm before the pressure threshold for motion is reached. This is a physical limitation of the actuator, not an artefact of the adaptation law.

7 Conclusion

This paper has presented a comprehensive MRAC design for a nonlinear EHA, identifying and resolving a fundamental structural limitation of the standard Lyapunov adaptation law for systems with $B = [0 \ 0 \ b_3]^\top$. Four contributions were established:

1. The dominance ratio $P_{33}/P_{13} = 771$ was shown to be intrinsic to the closed-loop Lyapunov equation and irreducible by LMI-based P -matrix optimisation.
2. The e_1 -projection was identified as output-feedback MRAC, providing a theoretically grounded justification for what appeared as a heuristic approximation.
3. Proposition 4.1 established that classical UUB via Young's inequality is structurally infeasible for this system class, and this constitutes an open theoretical problem.
4. Propositions 4.3–4.4 provide rigorous parameter-boundedness and practical-stability results under norm projection and σ -modification.

The design achieves RMS position error 0.45 mm (3 % of amplitude) with ≈ 5 s convergence time on the nonlinear model. Experimental validation and extension to online LuGre parameter identification are planned as future work.

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